

The influence of discourse context on children’s use of conversational devices

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BACKGROUND

- ❖ Certain conversational devices are useful in organizing discourse and ensuring mutual understanding, such as:
 - ❖ Backchannels (providing positive evidence; e.g., *mhm, uh-huh, yeah*)
 - ❖ Linguistic alignment (LA; matching interlocutor’s language):
 - ❖ Lexical alignment: **same word choice**
 - ❖ Syntactic alignment: same grammatical structure
 - ❖ Semantic alignment: same topic
- Child A: ***I think*** we should run away from ***this alien***.
Child B: Yeah, ***I think this alien*** is bad.

** these children are semantically aligning as they are talking about the same topic
- ❖ In adult-adult discourse, speakers adjust their uses of these devices depending on the context:^[1,2]
 - ❖ Task-oriented conversations (TOCs) =
 - ❖ higher levels of lexical & syntactic alignment
 - ↳ to increase precision to achieve shared goals^[1]
 - ❖ Spontaneously-occurring conversations (SOCs) =
 - ❖ more backchannels
 - ❖ higher rates of lexical & syntactic alignment
 - ❖ higher levels of semantic alignment
 - ↳ to maintain social relations by showing understanding and/or agreement^[1]
 - ❖ While this is the case for adults, **it remains unclear whether children are sensitive to the same contextual demands.**
 - ❖ Most studies looking at children’s use of these devices have been in the **context of adult-child discourse, where there is a clear asymmetry** (e.g., adult often leads discourse).^[3-6]
 - ❖ In these studies, children generally align less to adults than adults to children (e.g., lexical alignment rate: $M_{\text{child}}=0.23$; $M_{\text{adult}}=0.38$).^[3, 4]
 - ❖ Further, **other devices, such as discourse markers (DMs; e.g., *like, you know*) also support discourse organization,^[4] but have not been investigated extensively with regards to context.**

OBJECTIVE:

To determine whether school-aged children’s use of conversational devices with their same-age friend is sensitive to contextual demands

METHODS

- ❖ 30 English-speaking school-aged children (range=8-to-14 years; $M_{\text{age}}=9.7$) nested in 15 friendship dyads
 - ❖ Each child in a given dyad was matched on age (± 1 year) and expressive language abilities.
- ❖ Dyads engaged in two TOCs (untimed computerized and physical games that required joint decision-making) and two SOC (unstructured conversations).

LANGUAGE CODING

- ❖ Backchannel and DM tokens were identified via utterance-by-utterance coding.
- ❖ LA was calculated using an open-source Python program, ALIGN.^[6]
 - ❖ LA was calculated in terms of:^[3]
 - ❖ **Rate**: proportion of turns in which alignment occurred at all
 - ❖ **Level**: degree of alignment when alignment occurred

ANALYSES

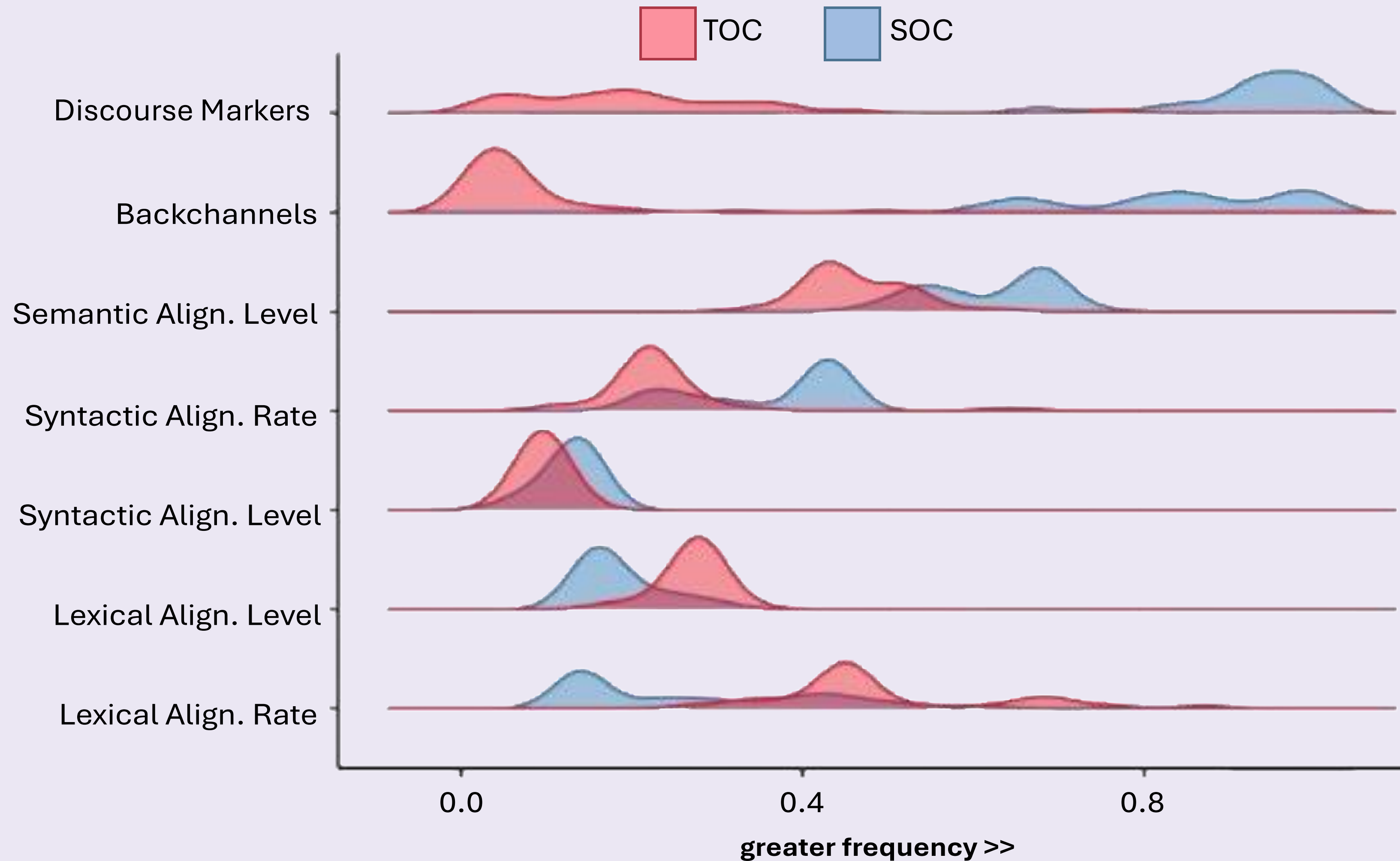
- ❖ To account for differences in verbosity, analyses were conducted on proportion of backchannel and DM tokens out of all utterances and alignment rates out of all conversational turns.
- ❖ Bayesian multilevel logistic regression models were built with each child nested in their respective dyad to determine children’s propensity to produce a particular device in a given context.
- ❖ The two TOCs and the two SOC were first combined into singular TOC v. SOC contexts (**Figure 1**), and then analyzed separately (**Table 1**).

DISCUSSION

- ❖ We replicated some patterns from the adult literature in our sample of school-aged children, suggesting that **different contexts afford diverse uses of conversational devices**, even for children.
 - ❖ Like adults, children may be using **lexical alignment to be precise and faster** (Figure 2) **to achieve shared goals in TOCs**, and using **backchannels, DMs, and syntactic and semantic alignment to maintain social relations in SOC**.
- ❖ Further, we found new context-specific patterns (i.e., greater lexical alignment level in TOCs; more DMs and higher rate of syntactic alignment in SOC).
 - ❖ This could stem from **the overall child-child discourse context being more symmetrical than adult-child discourse** (i.e., both children share authority and control over the discourse).
 - ❖ Rather than relying on an adult, **children in child-child discourse have free reign over their own linguistic choices, yielding greater uses of particular devices** than previously reported in the context of child-adult discourse, such as lexical alignment rate,^[3,4] and thus revealing new patterns.

RESULTS

Figure 1. Effect of TOC/SOC on conversational devices



NOTE: Each ridge plot represents the density of frequency values observed for each conversational device across the two types of contexts. Devices with peaks towards the left were less frequently used, while peaks to the right indicate greater frequency of use. The overlap between the blue and red distributions highlights similarities between SOC and TOC contexts, whereas distinct peaks or shifts suggest differences in device usage across contexts.

REPLICATED FINDINGS:

- ❖ TOCs:
 - ❖ higher level of lexical alignment
- ❖ SOC:
 - ❖ more backchannels
 - ❖ higher level of semantic alignment

NEW FINDINGS:

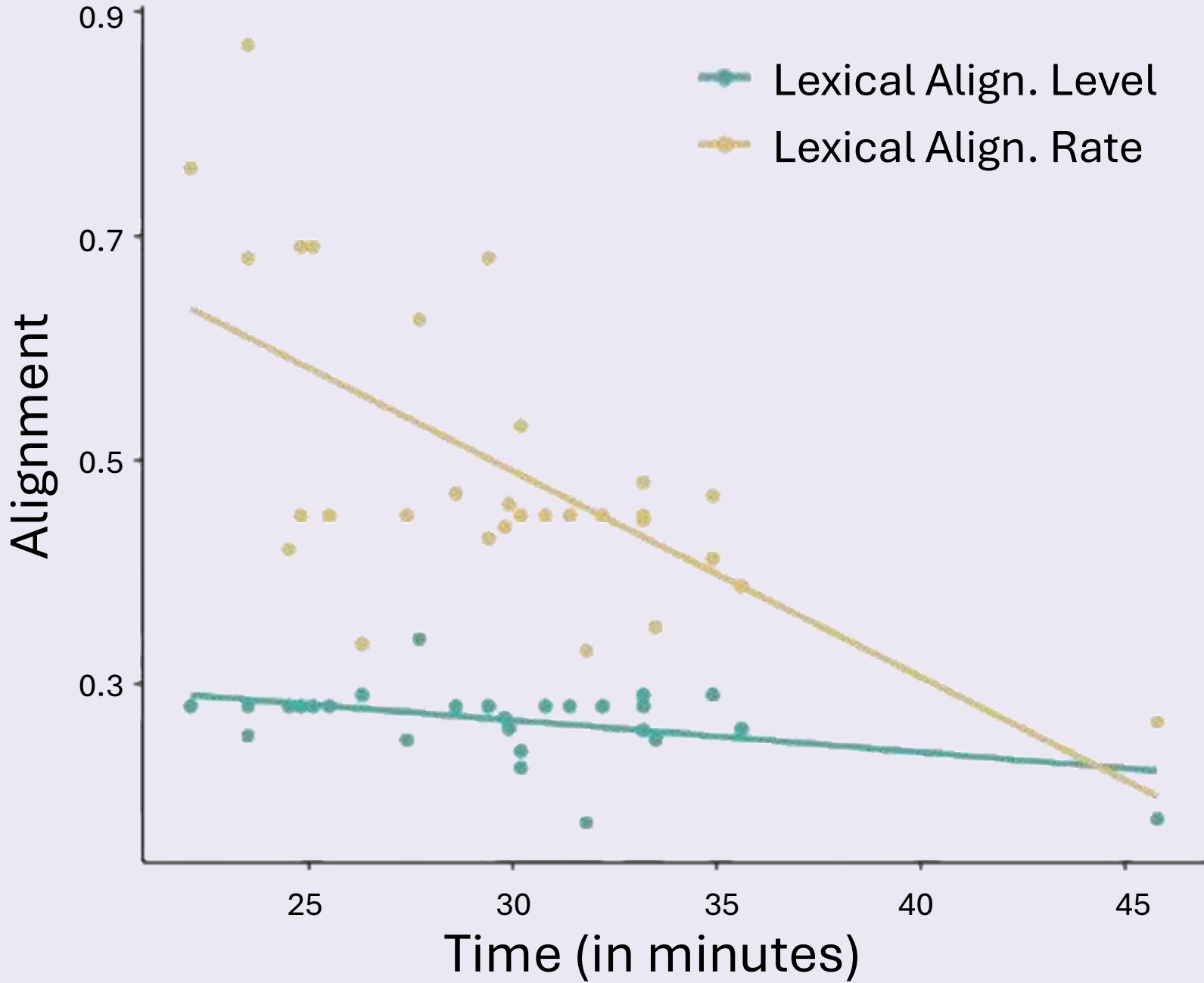
- ❖ TOCs:
 - ❖ higher rate of lexical alignment
- ❖ SOC:
 - ❖ more DMs
 - ❖ higher rate of syntactic alignment

Table 1. Posterior estimates for each conversational device for each context

	TOC 1 – Computer Game	TOC 2 – Physical Game	SOC 1	SOC 2
DMs	9.6%	20.8%	25.5%	30.9%
Backchannels	1.6%	3.4%	28.1%	36.9%
Semantic Level	18.8%	45.2%	56.1%	70.1%
Syntactic Rate	30.8%	37.6%	54.1%	62.3%
Syntactic Level	10.0%	9.8%	7.6%	8.8%
Lexical Rate	39.9%	37.8%	22.9%	14.3%
Lexical Level	37.9%	7.5%	22.9%	14.3%

NOTE: larger posterior estimates indicate more of that device in that context; bolded posterior estimate value indicates which activity had the greatest amount of that device across the contexts; syntactic alignment level did not significantly differ across contexts

Figure 2. Correlations between lexical alignment and time it took to complete the untimed games (i.e., TOCs)



Lexical alignment rate ($r = -0.647$) and level ($r = -0.436$) were **negatively correlated** to time ($ps < 0.05$).

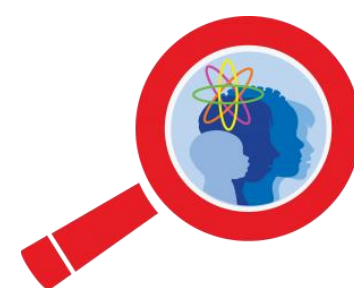
In other words,
greater lexical alignment → faster completion of games

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